OBSTACLE AVOIDANCE CAR

PROJECT NAME:	Obstacle Avoidance Car
PROJECT GROUP:	Gaurav Ingle

WHY THIS PROJECT ?

- Helps understand basic robotics and automation.
- Demonstrates automatic obstacle detection.
- Reduces chances of collision.
- Useful for autonomous robot applications.

INTRODUCTION

- The obstacle avoidance car is an autonomous robotic vehicle that uses an ultrasonic sensor to detect obstacles in its path and automatically changes direction with the help of an Arduino-controlled motor system.
- This project demonstrates the use of sensors, microcontrollers, and motor drivers to build a smart vehicle that can move safely without human control by avoiding collisions with nearby object.

KEY FEATURES

- Uses an ultrasonic sensor to measure distance and detect obstacles in front of the car.
- Arduino Uno controls the system and processes sensor data for decision making.
- Motor driver module controls the speed and direction of the DC motors.
- The car automatically changes direction when an obstacle is detected.

ARCHITECTURE

- The ultrasonic sensor continuously measures the distance between the car and nearby obstacles.
- The sensor sends distance data to the Arduino Uno for processing.
- The Arduino analyzes the data and decides the direction of movement.
- The motor driver module receives signals from the Arduino and controls the DC motors.
- The car moves forward or changes direction to avoid obstacles automatically.

RESULTS

- Obstacles were detected successfully.
- Car avoided collisions automatically.
- Smooth autonomous movement achieved.

FUTURE SCOPE

1. **Use advanced sensors for better detection**
2. • **Wireless control system**
3. **Application in smart robots and vehicles**